▲ **A GUN CREW AT BATTLE STATIONS**
To fire a naval cannon, the crew rammed a gunpowder cartridge and cannonball down the barrel, "ran" the gun out so the barrel protruded from the gun port, and ignited the charge.

◄ **THE BATTLE OF QUIBERON BAY**
In November, 1759, British Admiral Edward Hawke pursued a French invasion fleet of 21 ships into Quiberon Bay, off the coast of France. The 24 British ships of the line shattered the French navy, effectively ending its role in the Seven Years' War.

perspective, in 1805 an entire corps of Napoleon's army had only 40 cannon. Over time there were significant but limited technological improvements. A flintlock mechanism was first applied to naval cannon in 1745, but the slow-burning linstock (a forked staff that held a match) was still used in the early 19th century to fire cannon aboard many warships. The carronade, a light but powerful short-range gun introduced in the 1770s, was a notable addition to the firepower of naval fleets.

TECHNOLOGY AND TACTICS

More influential were general improvements in sailing technology, from accurate navigation to progress in nutrition and medical care. The practice of sheathing underwater wooden hulls with copper, widely adopted from the 1770s, made ships faster and able to stay at sea for longer periods without docking for careening (cleaning and repair). The chief developments, however, were tactical rather than technological. Britain's Royal Navy – by the early 17th century the world's leading naval power – traditionally had a commitment to attack and, with superior gunnery, expected to defeat its enemies if it could bring them to battle. In the Seven Years' War (1756–63), the boldest British admirals practised aggressive tactics known as the "general chase" – each ship racing to engage a fleeing enemy without waiting to take its allotted place in a formal line of battle. From the 1780s onwards, the British developed the tactic of cutting through the enemy line, instead of sailing parallel to it, so that part of the opposing fleet could be surrounded and destroyed. This strategy was taken to its extreme by Admiral Horatio Nelson, who sought to force the enemy into a "pell-mell" battle – a savage and chaotic close-quarters mêlée in which British gunnery and fighting spirit would carry the day. Nelson won decisive victories with these risky tactics and helped secure British naval dominance for another century.

> "No captain **can do very wrong** if he places his ship alongside that of **an enemy**"

ADMIRAL NELSON, BEFORE THE BATTLE OF TRAFALGAR, OCTOBER 1805

KEY **EVENTS**

1740–1805

- **1740** The French warship *Médée* is the first of a new type of frigate that will play a large part in naval warfare into the 19th century.
- **1761** On an experimental basis, the British Royal Navy coppers the hull of the 32-gun frigate HMS *Alarm*.
- **1763** The French navy adopts a system of signal flags that includes a number code. These greatly improve the visual communication of messages at sea.
- **1782** At Saintes, in the Caribbean, a British fleet under Admiral George Rodney breaks through the French battle line, initiating a new phase in British naval tactics.
- **1794** The US Navy is founded by an act of Congress, leading to the construction and manning of six powerful 44-gun or 38-gun frigates.
- **1805** Victory over a French and Spanish fleet at Trafalgar establishes a durable British command of the sea; Admiral Nelson of the British fleet is killed in the battle.